

ANCHORAGE MERRILL FIELD, AK, USA

WMO: 702735

Lat: 61.217N

Lon: 149.855W

Elev: 42

StdP: 100.82

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26409

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.3	-21.1	-26.7	0.3	-22.0	-24.2	0.4	-19.5	8.0	-1.2	6.9	-1.1	0.8	80	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	23.4	15.5	21.9	14.9	20.3	14.0	16.4	21.9	15.6	20.3	14.8	19.0	3.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.0	17.2	13.2	9.5	16.9	12.6	9.1	16.6	45.7	21.9	43.5	20.5	41.4	18.8	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	5.5	4.7	-25.5	26.3	2.8	2.0	-27.5	27.7	-29.1	28.9	-30.7	30.0	-32.8	31.5		
			-25.7	17.8	2.8	1.7	-27.7	19.1	-29.3	20.1	-30.9	21.0	-32.9	22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-8.8	-6.1	-3.3	3.8	9.9	14.2	16.1	14.7	10.1	2.7	-5.0	-7.7
	DBStd	10.07	7.07	5.84	5.12	3.48	2.80	2.41	2.11	2.15	2.86	4.51	5.73	5.99
	HDD10.0	2931	582	451	413	186	37	1	0	0	33	228	451	549
	HDD18.3	5449	841	684	671	436	263	125	75	113	248	486	701	808
	CDD10.0	533	0	0	0	1	32	128	189	147	36	1	0	0
	CDD18.3	11	0	0	0	0	0	2	6	2	0	0	0	0
	CDH23.3	53	0	0	0	0	1	16	27	9	0	0	0	0
	CDH26.7	4	0	0	0	0	0	2	2	0	0	0	0	0
Wind	WSAvg	1.7	1.4	1.4	1.8	1.9	2.2	2.2	2.0	1.7	1.6	1.6	1.5	1.4
Precipitation	PrecAvg	416	20	20	17	13	17	28	46	71	75	49	30	29
	PrecMax	700	69	61	70	48	49	86	114	248	196	109	72	68
	PrecMin	206	0	3	0	0	1	4	5	8	18	9	1	2
	PrecStd	89	14	12	13	12	11	19	25	45	44	24	21	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	6.9	8.6	15.0	22.5	25.5	26.2	24.6	18.9	13.3	8.6	8.0
		MCWB	3.1	2.9	3.8	8.0	12.1	15.9	16.8	16.4	12.3	8.5	4.2	3.6
	2%	DB	4.1	4.1	6.4	12.6	19.3	22.8	23.6	22.4	17.2	11.1	5.1	4.0
		MCWB	1.8	1.7	2.5	6.3	10.8	14.6	16.2	15.5	11.8	7.2	3.1	1.8
	5%	DB	2.1	2.5	4.8	11.1	17.2	21.0	22.0	20.6	15.9	9.7	3.6	1.7
		MCWB	0.7	1.1	1.5	5.5	9.7	13.8	15.3	14.6	11.1	6.6	2.0	0.5
	10%	DB	0.2	1.3	3.2	9.6	15.3	19.0	20.4	19.0	14.3	8.4	2.3	0.1
		MCWB	-0.8	0.0	0.7	4.7	8.8	12.8	14.6	13.9	10.6	6.2	1.0	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.2	4.6	8.9	12.9	16.9	17.8	17.3	13.4	9.7	4.8	3.8
		MCDB	6.5	5.4	7.7	14.0	21.1	24.1	24.8	23.2	16.6	11.7	7.4	7.1
	2%	WB	1.9	2.3	3.0	7.1	11.3	15.3	16.6	16.2	12.5	8.1	3.4	1.9
		MCDB	3.8	3.9	5.6	11.4	17.7	21.5	22.4	20.8	15.5	9.7	5.2	3.7
	5%	WB	0.6	1.2	1.9	5.9	10.3	14.2	15.8	15.3	11.9	7.3	2.2	0.5
		MCDB	2.1	2.5	4.2	10.2	16.0	19.5	20.7	19.3	14.5	9.0	3.4	1.5
	10%	WB	-0.8	0.1	1.0	5.0	9.4	13.4	15.1	14.6	11.3	6.4	1.0	-0.7
		MCDB	0.2	1.3	3.0	9.1	14.5	18.3	19.4	18.1	13.7	8.3	2.1	0.1
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.3	8.8	8.6	8.6	7.9	7.0	7.1	7.0	6.2	5.9	6.2
		MCDBR	7.0	7.0	8.4	10.8	11.8	10.9	10.0	10.0	8.9	6.9	5.9	7.4
		MCWBR	5.1	5.1	5.5	6.1	6.0	5.4	4.9	4.9	4.9	4.2	4.3	5.6
	5% WB	MCDBR	6.8	6.6	7.5	9.7	10.2	9.7	9.0	8.4	7.3	5.7	5.5	6.8
		MCWBR	5.1	5.0	5.1	5.9	5.5	5.0	4.5	4.6	4.3	3.7	4.1	5.3
Clear-Sky Solar Irradiance	taub	0.233	0.286	0.305	0.346	0.367	0.358	0.371	0.356	0.317	0.296	0.241	N/A	
	taud	2.128	2.233	2.267	2.248	2.314	2.421	2.415	2.463	2.558	2.471	2.126	N/A	
	Ebn at Noon	505	694	813	843	853	870	847	832	812	696	473	N/A	
	Edh at Noon	37	65	89	111	113	103	102	90	68	52	36	N/A	
All-Sky Solar Radiation	RadAvg	0.29	0.97	2.59	4.29	5.16	5.32	4.73	3.74	2.44	1.13	0.42	0.15	
	RadStd	0.04	0.14	0.28	0.38	0.46	0.34	0.41	0.41	0.34	0.14	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	+1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	+2.00	+1.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page